Section 2.1 — Monomials and Polynomials — Answer Key

Math 103: College Algebra

1 Exponents

For each expression below:

a. Identify the base and the exponent.

b. Rewrite as repeated multiplication.

c. Evaluate by multiplying, then check your answer using the exponent key on your calculator.

1. a. Base: 2, Exponent: 3 b. $2 \cdot 2 \cdot 2$ c. 8
2. a. Base: 3, Exponent: 4 b. $3 \cdot 3 \cdot 3 \cdot 3$ c. 81
3. a. Base: -4, Exponent: 2 b. $(-4)(-4)$ c. 16
4. a. Base: 5, Exponent: 3 b. $5 \cdot 5 \cdot 5$ c. 125
5. a. Base: -1, Exponent: 5 b. $(-1)(-1)(-1)(-1)(-1)$ c. -1

For each of the expressions below:

a. Rewrite using a positive exponent.

b. Evaluate using your rewritten expression, then check your answer using your calculator. Answer as a fraction.

6. $\frac{1}{3^2} = \frac{1}{9}$
7. $\frac{1}{5^1} = \frac{1}{5}$
8. $\left(\frac{3}{2}\right)^3 = \frac{27}{8}$
9. $\frac{1}{(-4)^3} = -\frac{1}{64}$
10. $\left(-\frac{1}{3}\right)^4 = \frac{1}{81}$

2 Simplifying Monomials

Simplify each monomial below, applying exponent rules as appropriate.

11. b^5
12. $10y^5z$
13. $9x^6$
14. $\frac{81}{16}x^4$
15. $64x^{14}y^2$
16. $\frac{1}{6}x^3y^3$
17. $-84h^3j^5k^2y$

3 Using Polynomial Vocabulary

For each expression, decide whether it is

A. not a polynomial B. a monomial C. a binomial

D. a trinomial E. a polynomial with 4 or more terms

18. C
19. B
20. B
21. D
22. B
23. B
24. A
25. A
26. B
27. B
28. E
29. A
30. B
31. B
32. A
33. C

For each polynomial below:

a. Identify the leading term and leading coefficient, if any.

b. Give the degree of the polynomial.

c. Identify any that are constant, linear, quadratic, or cubic.

34. a. Leading term: N/A, Ldg. Coefficient: N/A b. Degree: 1 c. Linear

35. a. Leading term: $17.3x$, Ldg. Coefficient: 17.3 b. Degree: 2 c. Quadratic

36. a. Leading term: $18xy^3$, Ldg. Coefficient: 18 b. Degree: 4 c. None of these

37. a. Leading term: x^3, Ldg. Coefficient: 1 b. Degree: 3 c. Cubic

38. a. Leading term: $7k$, Ldg. Coefficient: 7 b. Degree: 1 c. Linear

39. a. Leading term: 36, Ldg. Coefficient: 36 b. Degree: 0 c. Constant

40. a. Leading term: $-x^2$, Ldg. Coefficient: -1 b. Degree: 2 c. Quadratic

41. a. Leading term: $-u^7$, Ldg. Coefficient: -1 b. Degree: 7 c. None of these

Give an example of each of the following.
Answers will vary; these are possible examples.

42. $3x^2 - 5x + 2$

43. $x^3 + x^2y + xy^2 + y^3$

44. $2x - 5$

45. $x^3 + 8$

4 Polynomial Functions

Let $f(n) = n^2 - 8n - 3$. Find:

46. $f(2) = -15$ 47. $f(-3) = 30$ 48. $f(\frac{1}{2}) = -\frac{27}{4}$ 49. $f(0) = -3$

Let $g(j) = 22j^3 + 5j^2 - 16j^4 + 21 - 14j$. Find:

50. $g(1) = 18$ 51. $g(-2) = -363$ 52. $g(\frac{1}{4}) = 579/32$ 53. $g(-1) = 2$